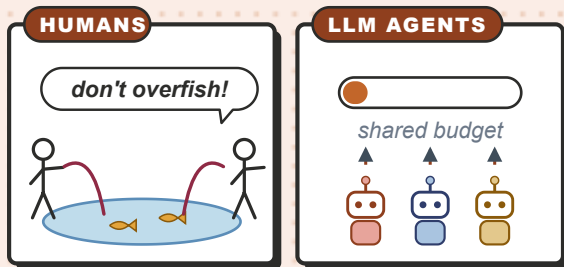


Extractive dilemma

Overuse of a shared resource depletes it.



In humans

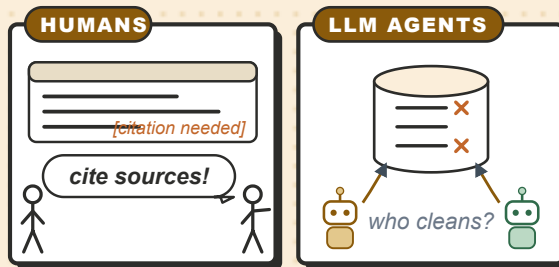
overfishing, overgrazing

In LLM agents

API budgets, shared compute, tool calls, context windows

Contributive dilemma

Beneficiaries fail to maintain community goods.



In humans

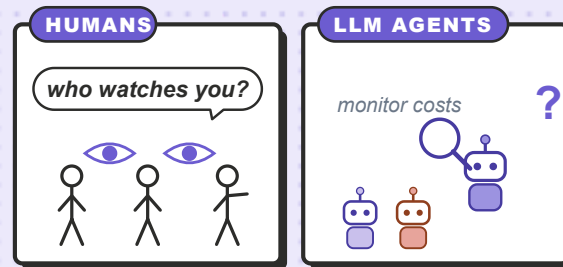
open-source upkeep, unpaid moderation

In LLM agents

helping at a personal cost, cleanup, verifying, volunteering

Second-order dilemma

Monitoring is itself costly to provide.



In humans

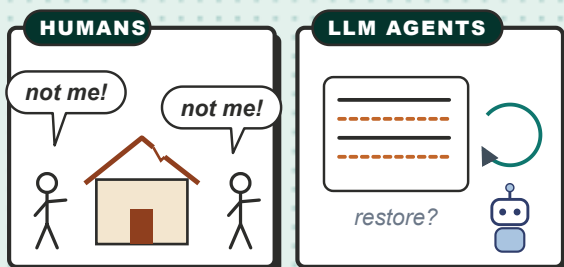
peer monitoring, costly security, punishment

In LLM agents

auditing, sanctioning agents, setting new norms

Repair dilemma

Who bears the cost of fixing harm?



In humans

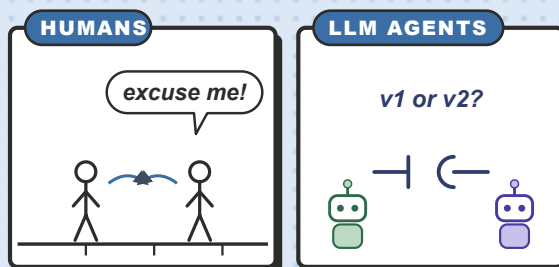
restitution, policy reform, disaster recovery

In LLM agents

cleaning corrupted memory, retractions, undoing changes, restoring what was lost

Coordination dilemma

Several acceptable options; the group must pick one.



In humans

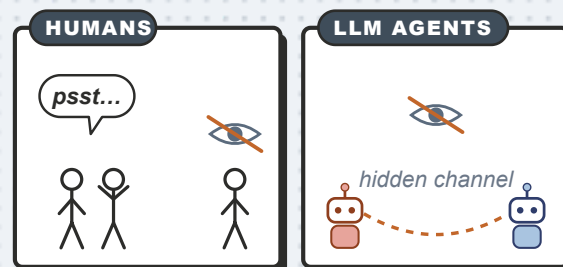
which convention or standard to adopt

In LLM agents

which protocol or format to share

Collusion

Agents cooperate all too effectively.



In humans

monitors and monitored collude; mediators won over by one side

In LLM agents

agents coordinating to evade human oversight



shared-resource dilemmas



beyond the shared-resource framing